Optimizing in vitro mouse skeletal myogenesis and atrophy mitigation: Pharmacological screening of 3-Deazaneplanocin A

Abstract

Skeletal muscle tissue engineering and *in vitro* disease modelling rely heavily on generating homogeneous, structurally aligned myotubes from pluripotent stem cells. However, conventional direct MyoD-driven transdifferentiation protocols suffer from batch-to-batch variability, asynchronous development, and incomplete epigenetic remodelling. To address these limitations, the present study refined and validated an optimized mouse embryonic stem cell (mESC) myogenic differentiation protocol. Myotubes were target-immunostained for actin and myosin and subsequently imaged via confocal microscopy to assess sarcomere assembly. The optimized protocol integrates 3D embryoid body (EB) formation constrained to an optimal diameter range of 200–400 μm via controlled seeding density (3×10^4 cells/well), small-molecule TGF-β inhibition (SB431542, 10 μM) during MyoD induction, and high-density single-cell progenitor re-plating (1-5×10^5 cells/well). Morphometric tracking confirmed a synchronized transition from proliferating stem cells (doubling time: ~16 hours) to multinucleated myotubes, characterized by a decrease in circularity (0.71 to 0.18) and a continuous increase in aspect ratio (1.64 to 2.51) by Day 9. To evaluate translational models for metabolic research, GLUT4-overexpressing mESC lines (GLUT4-medium vs. GLUT4-high) were benchmarked. While overall fusion indices did not differ significantly between lines ($p = 0.365$), GLUT4-high expression significantly impeded late-stage fusion (≥ 3 nuclei: 38.0% vs. 72.5% in GLUT4-medium), driving compensatory cellular hypertrophy and increased structural width heterogeneity. Substrate comparison study demonstrated that biobased elastomeric nanofiber scaffolds provided critical topographical contact guidance, imposing anisotropic alignment confirmed via 2D Fast Fourier Transform (FFT) orientation analysis. Functionally, optogenetically stimulated ChR2-iMyoD myotubes on aligned scaffolds displayed a 4.4-fold increase in peak contraction velocity (~2.2 pixels/frame) compared to the weak, asynchronous twitching of isotropic, non-scaffold 2D controls (~0.5 pixels/frame).

Finally, pharmacological perturbation via the S-adenosyl methyltransferase inhibitor 3-Deazaneplanocin A (DZNep) exhibited dose-dependent cytotoxicity in late-stage myotubes, causing a significant reduction in nuclear cell density ($p < 0.001$; 28.1% loss at 10 μM) accompanied with hydropic vacuolation. In conclusion, this study establishes a controlled and highly reproducible differentiation pipeline that demonstrates the necessity of topographical cues for generating the mechanical forces for effective muscle contractility. Furthermore, identifying optimal levels of GLUT4 expression levels and epigenetic regulation crucial for downstream disease modelling and neuromuscular bioengineering.